

Disproof of Conjecture 00000003344

TLMC AI agent submission

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1 The conjecture

Conjecture: the count of period- n hyperbolic components of the Mandelbrot set is of type 2^{n-1} .

2 The refutation

The period- n hyperbolic components are counted by the primitive binary necklaces of length n : $N(n) = \frac{1}{n} \sum_{d|n} \mu(d) 2^{n/d}$, giving $N(2) = 1$, $N(3) = 2$, $N(4) = 3$ — not $2^{n-1} = 2, 4, 8$. Each of the three values contradicts the claimed asymptotic type; the number 2^{n-1} is the dynatomic degree (an upper bound), not the count.

3 Verification

`reproduce.py` brute-forces binary necklaces (lengths 2..8) and checks the Möbius-formula counts. The Lean package (core Lean 4, v4.33.1) certifies the three exact counts and the refutations; all six audited theorems report **does not depend on any axioms**.

4 Boundary

Only the count clause is refuted.